

Control Architectures and Design Example

R0005E - Measurement and Feedback Control

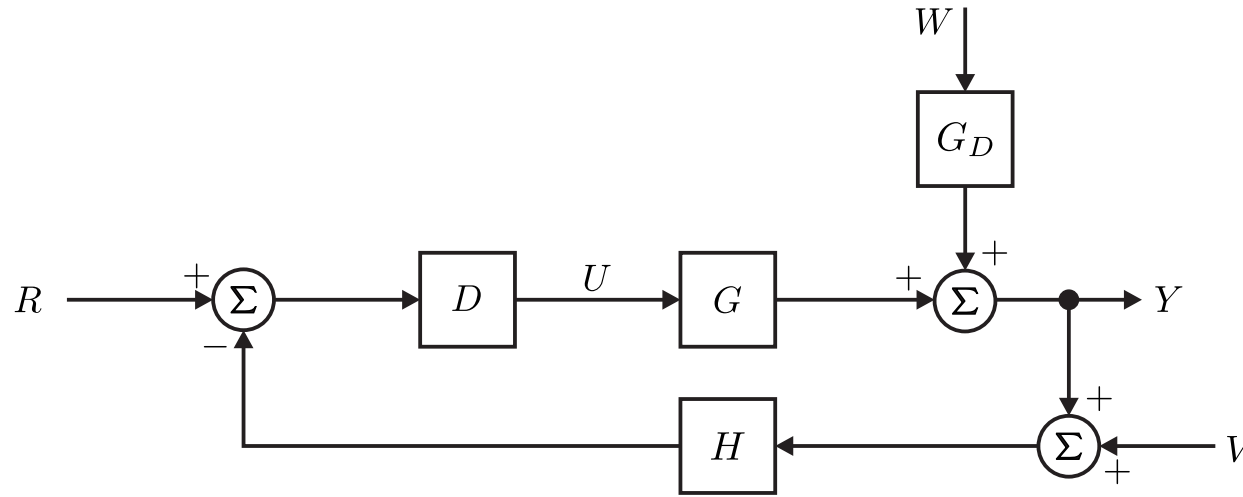
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General 1-DOF architecture

Until now we have studied the so-called one degree of freedom control (1-DOF) architecture.



We know that the closed loop transfer function from $R(s)$ to $Y(s)$ is given as follows:

$$T(s) = \frac{Y(s)}{R(s)} = \frac{G(s)D(s)}{1 + H(s)G(s)D(s)}$$

For now we will assume that $H(s) = 1$ and that there is no noise $V(s)$ present. Moreover, we combine $L(s) = G(s)D(s)$ and will call this transfer function the **loop gain**.

In the pole placement approach and the traverse crane example we have seen that there are zeros in $T(s)$, which could lead to undesired behaviour of the closed loop, like for example overshoots.

Assume:

$$L(s) = \frac{N(s)}{M(s)}$$

Where $N(s)$ is the numerator polynomial containing all the zeros and $M(s)$ is the denominator polynomial containing all the poles $M(s)$. Then we can rewrite $T(s)$ to

$$\begin{aligned} T(s) &= \frac{\frac{N(s)}{M(s)}}{1 + \frac{N(s)}{M(s)}} \\ &= \frac{N(s)}{N(s) + M(s)} \end{aligned} \quad (1)$$

Thus, all the zeros in the controller $D(s)$ and the process $G(s)$ are also the zeros of the closed loop system. When designing the controller to achieve a desired characteristic polynomial, there is no freedom left to design the zeros of the closed loop system.

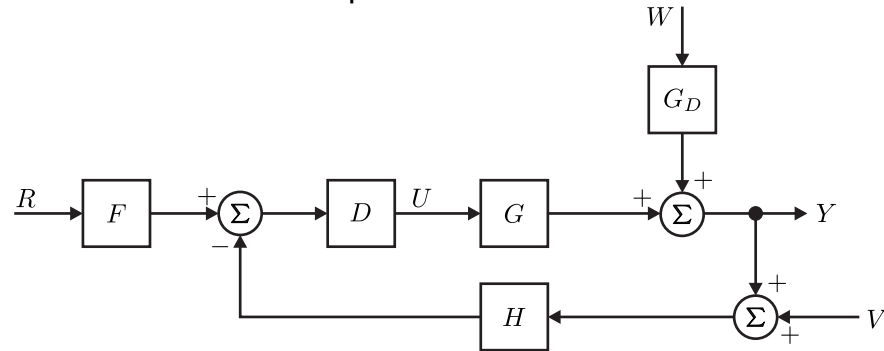
There are now two solutions:

- Choose the controller such that $L(s) = \frac{\lambda}{s}$
This approach is common in so-called loop shaping or internal model control (IMC).
- Make use of a pre-filter $F(s)$
This approach is a so-called 2-DOF architecture.

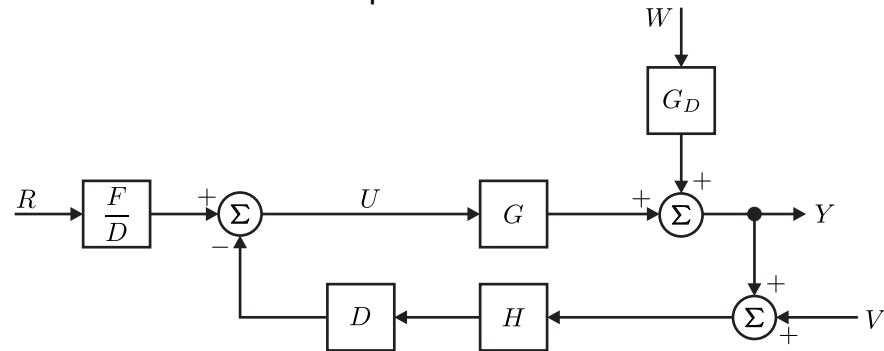
General 2-DOF architecture

The two degree of freedom architecture can be given in two forms:

- Controller in forward path



- Controller in feedback path



In both cases there is a filter in the forward path outside the feedback loop, which can be used to shape the zeros of the closed loop system. Depending on implementation issues, one of the above architectures is chosen.

Remember, that the components of control system need to be proper to be implementable.

In general, the 2-DOF architecture enables the engineer to shape the regulation and tracking properties of the closed loop system independently.

Now, we can derive the closed loop transfer function again. We also assume that there is no noise $V(s)$ present and that $H(s) = 1$.

$$\begin{aligned} T(s) &= F(s)T'(s), \quad \text{with} \quad T'(s) = \frac{N(s)}{N(s) + M(s)} \\ &= \frac{F(s)N(s)}{N(s) + M(s)} \end{aligned} \tag{2}$$

In (2), we can see that $F(s)$ can be used to compensate for the zeros of the closed loop, but also see to that the DC-Gain of the closed loop system becomes 1. Assuming $T(0) = 1$, yields

$$F(0)T'(0) = 1 \quad \Rightarrow \quad F(0) = \frac{1}{T'(0)}$$

Note:

- The pre-filter $F(s)$ can be designed after the closed loop is designed. This means one can test and implement $F(s)$ and $D(s)$ independently.
- The pre-filter $F(s)$ can be used to compensate for steady state control errors in the tracking properties of $T'(s)$.

Control action in 2-DOF

As in all closed systems the behaviour of the control action is important to understand. This is especially true in the 2-DOF architecture. Let's derive the control action for the 2-DOF architecture

$$\begin{aligned}U(s) &= D(s) \{F(s)R(s) - H(s)(G_D(s)W(s) + G(s)U(s))\} \\&= D(s)F(s)R(s) - D(s)H(s)G_D(s)W(s) - D(s)H(s)G(s)U(s) \\ \Rightarrow \quad U(s) &= \frac{D(s)F(s)}{1 + D(s)H(s)G(s)}R(s) - \frac{D(s)H(s)G_D(s)}{1 + D(s)H(s)G(s)}W(s)\end{aligned}$$

Note:

- The gain from the exogenous signal signals $R(s)$ and $W(s)$ to the control action $U(s)$ has to be limited such that actuator saturations do not or only rarely occur.
- Based on an understanding of the process, the typical signal size or magnitude of $R(s)$ and $W(s)$ can be judged. At the same time the actuator limits of $U(s)$ are known. Jointly, this knowledge gives an understanding of the admissible gain of the transfer functions for $U(s)/R(s)$ and $U(s)/W(s)$.
- $F(s)$ can also be used to shape the transfer function from $R(s)$ to $U(s)$, but not from $W(s)$ to $U(s)$.

Improving the disturbance rejection

We have seen that the regulation properties are not affected by the pre-filter $F(s)$, which means that the disturbance rejection is not improved.

Idea: Make use of information on the disturbance and use it in the disturbance rejection.

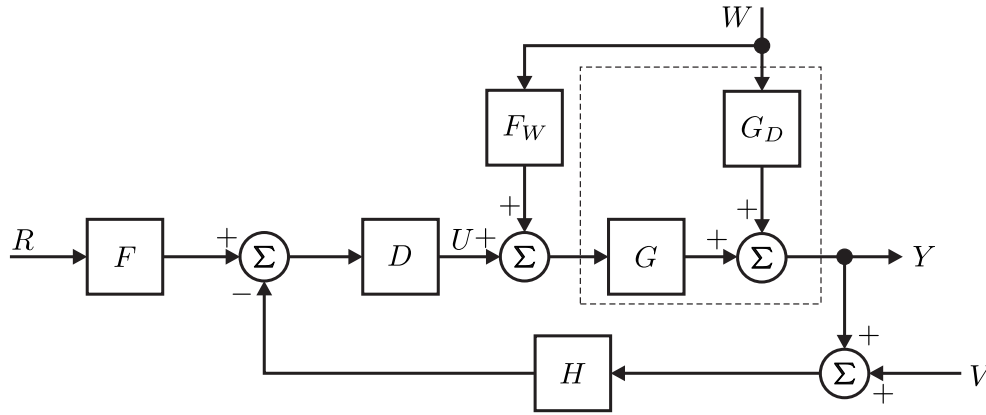
If there is additional information available, this information can be used to improve the regulation properties.

There are now two cases:

- The disturbance can be measured.
Make use of the measurement and counteract the disturbance effect. This can be achieved by a so called **feedforward architecture**.
- Another measurement is affected much earlier by the disturbance.
Build an additional feedback structure which counteracts the disturbance effect early. This can be achieved by a so-called **cascade architecture**.

Disturbance rejection with feedforward

Principle sketch of the architecture:



Now that $W(s)$ can be measured, a feedforward transfer function $F_W(s)$ can be designed which counteracts the disturbance effect on $Y(s)$. The output $Y(s)$ can be written as

$$\begin{aligned} Y(s) &= G_D(s)W(s) + G(s)(U(s) + F_W(s)W(s)) \\ &= G_D(s)W(s) + G(s)F_W(s)W(s) + G(s)U(s) \end{aligned}$$

The disturbance effect is completely compensated for when $G_D(s)W(s) + G(s)F_W(s)W(s) = 0$. This then leads to

$$\Rightarrow G_D(s) + G(s)F_W(s) = 0$$

$$\Rightarrow F_W(s) = -\frac{G_D(s)}{G(s)}$$

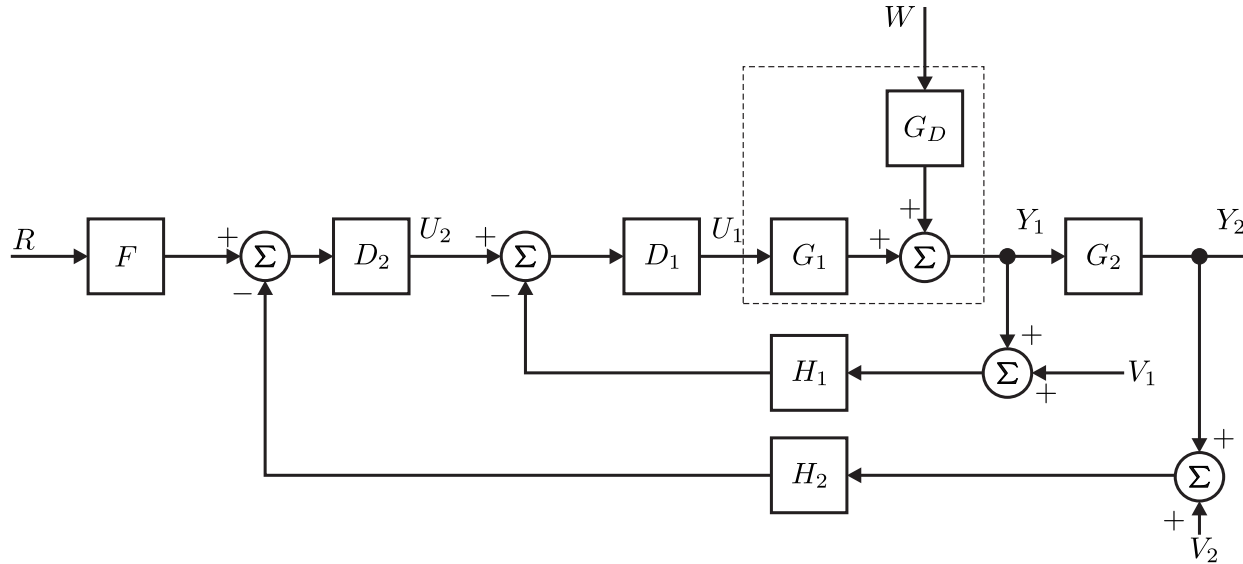
In order to implement such a feedforward transfer function, this transfer function need to be proper. In many cases this is not true and approximation has to be found. One approximation is the steady state compensation which is given by

$$F_W(0) = -\frac{G_D(0)}{G(0)}$$

Later in the course, we will design simple filters, which can also be used to approximate the feedforward transfer function.

Disturbance rejection with a cascade

Principle sketch of the architecture:



In this case, the disturbance $W(s)$ is entering before the measured signal $Y_1(s)$, which is well before the measured signal $Y_2(s)$. Designing an inner loop using the real actuator $U_1(s)$ will yield a disturbance rejection of $W(s)$, while the outer loop will take care of the tracking properties.

It has to be noted that $U_2(s)$ is not not a real actuator, but the reference signal to the inner loop. The outer loop may also deal with disturbances which are not reflected in $Y_1(s)$.

Note:

- We have to design two controllers $D_1(s)$ and $D_2(s)$.
- The two controllers depend on each other, but we want to design the controllers independently. For this reason we design the inner loop fast enough, so that it reaches the steady state well before the outer loop acts. Essentially, the inner loop should reach steady state fast.

The closed loops can now be stated as follows, assuming $V_1(s) = V_2(s) = 0$:

Inner loop:

$$Y_1(s) = \underbrace{\frac{G_1(s)D_1(s)}{1 + H_1(s)G_1(s)D(s)}}_{T_1(s)} U_2(s) + \underbrace{\frac{G_D(s)}{1 + H_1(s)G_1(s)D(s)}}_{G_D(s)S_1(s)} W(s)$$

Outer loop:

$$Y_2(s) = \frac{G_2(s)T_1(s)D_2(s)F(s)}{1 + G_2(s)T_1(s)D_2(s)} R(s) + \frac{G_2(s)G_D(s)S_1(s)}{1 + G_2(s)T_1(s)D_2(s)} W(s)$$

If we now achieve that the inner loop reaches steady state fast with $S_1(0) = 0$ and $T_1(s) = 1$, then $Y_2(s)$ is given by

$$Y_2(s) = \frac{G_2(s)D_2(s)F(s)}{1 + G_2(s)D_2(s)} R(s)$$

Guideline

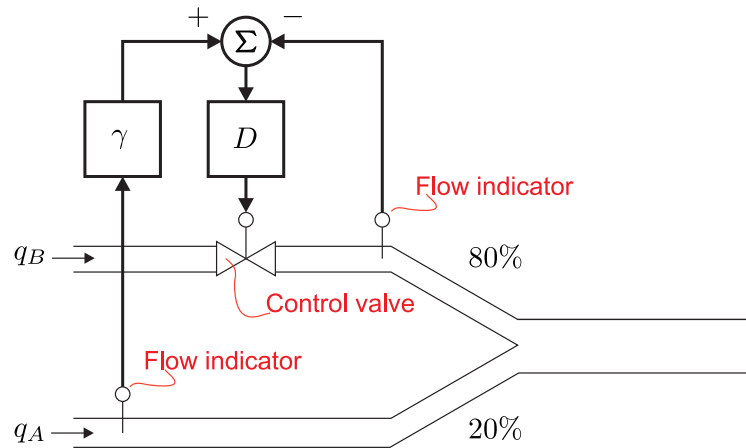
- The inner loop should be at least 6 times faster than the outer loop.
- If $S_1(0) = 1 - a$ and $T_1(0) = a$ with a small, then the gain a can be used in the design of $D_2(s)$ by combining it with $G_2(s)$.

Note: In the tank process it is possible to measure both upper tank level and lower tank level. This enables the design of a cascade architecture.

Ratio control

In many applications the ratio between two physical properties need to be kept constant. This could be for example the blend of two fluids, or the angular velocity of two wheels.

The problem can be solved by using one of the measured variables as the reference signals for the control loop of another variable. This architecture can be represented as follows



In this case, the two fluids should be blended at the ratio 80% to 20%. By adjusting the flow q_B based on the measured flow q_A it is possible to maintain a certain ratio which is set by the gain γ . The flow control loop for q_A can be designed independently.

Note: When ratio control is applied, the choice of the inner control loop is important. Always choose the loop which is easy and fast to control as the inner loop and the slow variable as the outer loop.

Design example - Traverse crane

We will now have a look at a traverse crane from earlier lectures and will design a closed loop system that fulfills certain specifications.

Remember:

$$G(s) = \frac{1.5}{s^2 + 0.4s + 1.5}, \quad G_w = \frac{2}{3}$$

Specification for the closed loop system:

- No overshoot for steps in the reference signal
- Settling time improved by 20%
- Affected of a constant wind disturbance completely attenuated

Procedure:

1. Analysis of the specification and determining the admissible pole region
2. Design of the closed loop system
3. Synthesis of the controller
4. First analysis of the closed loop system
5. Pre-filter design
6. Second analysis of the closed loop system